

Benjamin Carver

✉ benrcarver@gmail.com
🐙 github.com/scusemua
🏠 scusemua.github.io
📞 571 535 7343

Summary

- PhD in computer science with deep experience designing, developing, and evaluating large, scalable, high-performance distributed systems across multiple cloud platforms.
- Adaptive learner and adept problem-solver with proficiency in a wide array of programming languages, development tools, frameworks, and cloud services.
- Established, facilitated, and lead multiple collaborations with leading academic and industry partners.
- Motivated and driven to leverage strong academic and development background, along with substantial research expertise in distributed systems, cloud and serverless computing, and AI/ML, to help organizations build innovative, scalable, and high-performance infrastructure, products, and services.

Education

George Mason University, Fairfax, Virginia

- August 2025 *Computer Science*, Doctor of Philosophy, *Summa cum laude* (GPA: 4.0).
May 2021 *Computer Science*, Master of Science, *Summa cum laude* (GPA: 4.0).
May 2020 *Computer Science*, Bachelor of Science, *Summa cum laude* (GPA: 4.0).

Employment

- August 2025–Ongoing **Research Scientist**, META, Manhattan, New York, USA.
- Member of the Network.AI team, developing networking infrastructure for training large-scale AI models.
 - Improved initialization architecture and algorithms, cutting training job setup latency by 96% at multi-thousand-GPU scales, directly reducing startup and recovery times across Meta's GPU fleet.
 - Designed core low-level pipelining primitives and broadcast algorithms that are $2.4\times$ faster than NCCL at small messages, now foundational to Meta's next-generation collective protocols.
- Spring 2019–August 2025 **Graduate Research Assistant**, GEORGE MASON UNIVERSITY, Manassas, Virginia, USA.
- Served as the sole developer and tester of 3 complete, large-scale open-source distributed systems and as a co-developer of 2 other distributed applications of similar scale.
 - Experience creating and deploying distributed applications and services in cloud environments, including AWS, GCP, Microsoft Azure, and IBM Cloud, using Kubernetes and Docker (Compose/Swarm).
 - Published 5 academic papers in top-tier academic conferences, 4 of which were first-authorship publications, with 1 additional first-authorship paper close to submission.
 - Established, lead, and facilitated multiple collaborations with leading academic and industry research labs.
- May–August 2022 **Microsoft Research Intern**, MICROSOFT RESEARCH, Redmond, Washington, USA.
- Responsible for designing, developing, and evaluating a new microsecond-latency FaaS platform.
 - Improved API usability and reduced overheads by 25% through re-implementing client-facing interfaces as CPython extension modules, enabling multi-threading support and improving client application performance.
- June 2016–July 2020 **Software Developer**, BRTRC FEDERAL SOLUTIONS, Vienna, Virginia, USA.
- Optimized a .NET-based Geographic Information Systems (GIS) application, reducing load time by $5\times$ and memory footprint by 70%, significantly improving usability and the overall user-experience.

Publications

★: Tier-1 venue.

- [ASPLOS '26]★ **Benjamin Carver, Jingyuan Zhang, Haoliang Wang, Kanak Mahadik, Yue Cheng,**
NotebookOS: A Notebook Operating System for Interactive Training with On-Demand GPUs,
ACM International Conference on Architectural Support for Programming Languages and Operating
Systems (AR: 20/208 = 9.61%).
- Designed, implemented, and evaluated NotebookOS, a large-scale, fully-functional, GPU-efficient Jupyter Notebooks-as-a-Service platform designed for a unique class of deep learning workloads.
 - Designed and implemented a novel, kernel-replica-based design that saves 1,187+ GPU hours over a 17.5-hour real-world production workload while greatly enhancing user interactivity.
 - Enabled detailed, real-time metric and log collection, visualization, and monitoring by integrating multiple industry-standard frameworks and technologies, including Prometheus, Promtail, Loki, and Grafana.
 - Created robust, detailed simulation software to evaluate the pricing/cost, scheduling, fault tolerance, and scaling algorithms and policies of NotebookOS at extremely large scales.

- [ASPLOS '23]★ **Benjamin Carver, Runzhou Han, Jingyuan Zhang, Mai Zheng, Yue Cheng,**
λFS: A Scalable and Elastic Distributed File System Metadata Service using Serverless Functions,
ACM International Conference on Architectural Support for Programming Languages and Operating
Systems (AR: 151/600 = 25.16%).
- Designed, implemented, and evaluated λFS, a first-of-its-kind serverless metadata service for large-scale distributed file systems like Hadoop Distributed File System (HDFS).
 - Designed and implemented elastic auto-scaling mechanisms, efficient client libraries, and a novel FaaS-based metadata cache to provide 4.13× higher throughput, 90.40% lower latency, and 85.99% lower user-side cost compared to state-of-the-art baselines for real-world industrial workloads.
- [VLDB '23]★ **Jingyuan Zhang, Ao Wang, Xiaolong Ma, Benjamin Carver, Nicholas John Newman, Ali Anwar, Lukas Rupprecht, Vasily Tarasov, Dimitrios Skourtis, Feng Yan, Yue Cheng,**
InfiniStore: Elastic Serverless Cloud Storage, International Conference on Very Large Databases
(Average AR = 18.6%).
- Team member assisting in the testing and evaluation of INFINISTORE, a first-of-its-kind serverless-function-based data storage service that achieves 97.24% user-side cost savings compared to AWS ElastiCache.
 - Implemented a lightweight, microservice-based workload driver in Golang to execute large-scale, customizable MapReduce workloads as part of the evaluation of INFINISTORE and its predecessor, INFINICACHE.
- [SOCC '20]★ **Benjamin Carver, Jingyuan Zhang, Ao Wang, Ali Anwar, Panruo Wu, Yue Cheng,**
Wukong: A Scalable and Locality-Enhanced Framework for Serverless Parallel Computing,
ACM Symposium on Cloud Computing (AR: 35/143 = 24.4%).
- Designed, implemented, and evaluated WUKONG, a fully-featured serverless-computing-based execution engine created atop AWS Lambda and AWS Fargate.
 - Developed novel scheduling algorithms that use serverless functions to execute large-scale parallel computing jobs up to 68.17× faster while reducing user-side cost by 92.96% compared to state-of-the-art baselines.
- [PDSW '19] **Benjamin Carver, Jingyuan Zhang, Ao Wang, Yue Cheng,**
In Search of a Fast and Efficient Serverless DAG Engine,
International Parallel Data Systems Workshop.
- Created and evaluated system prototype that uses AWS Lambda to improve the performance of real-world data analytics, linear algebra, and AI/ML workloads by up to 3.1× compared to a non-serverless baseline.

Ongoing Research

A Microsecond-Scale Functions-as-a-Service (FaaS) Platform

- Team member responsible for designing, implementing, and evaluating a novel FaaS platform created to support modern scientific and data-analytics-based workloads.
- Integrated support for latest networking and virtualization technologies to enable orders-of-magnitude reduction in overhead incurred by the serverless platform.

Enhancing FaaS-Based Storage Systems for Privacy & Security

- Contributed to the design of a secure FaaS-based storage system that incorporates techniques from cryptography, network security, and database systems to provide strong security guarantees.

Skills

Languages	Python, Golang, Java, C, C++, C#, JavaScript/TypeScript, CSS.
Development Tools	VS Code, Visual Studio, JetBrains IDEs (IntelliJ/GoLand/WebStorm), Linux, Microsoft Office, Git (GitHub/GitLab/BitBucket), Jupyter Notebooks.
Frameworks & Services	Kubernetes, Docker, Docker Compose, Docker Swarm, Prometheus, Grafana, Loki, Promtail, HDFS, Dask, Pandas, NumPy, SciPy, PyTorch, JupyterLab, React, ZooKeeper, Redis, MySQL, Ansible.
Cloud	Amazon Web Services (AWS) EC2, VPC, Lambda, S3, IAM, CloudWatch, Fargate, ECS, EKS. Google Cloud Platform (GCP) Compute Engine, Google Kubernetes Engine (GKE), Cloud Functions. Microsoft Azure Cloud, IBM Cloud

Honors & Awards

May 2026	Distinguished Academic Achievement Award.
August 2023	VLDB2023 NSF Travel Grant.

- May 2021 **George Mason University Presidential Scholarship Award.**
Renewed annually for 4 years.
- May 2021 **Distinguished Academic Achievement Award.**
- May 2020 **Distinguished Undergraduate Research Award.**
- May 2020 **Distinguished Academic Achievement Award.**

Professional Services

External Reviewer

- 2026 **FAST**, *Fall Cycle*, 24th USENIX Conference on File and Storage Technologies.
- 2026 **EuroSys**, *Spring Cycle*, ACM European Conference on Computer Systems.
- 2026 **FAST**, *Spring Cycle*, USENIX Conference on File and Storage Technologies.
- 2025 **SC**, ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis.
- 2025 **HotStorage**, ACM Workshop on Hot Topics in Storage and File Systems.
- 2025 **ATC**, USENIX Annual Technical Conference.
- 2025 **FAST**, USENIX Conference on File and Storage Technologies.
- 2025 **HPDC**, ACM International Symposium on High-Performance Parallel and Distributed Computing.
- 2025 **NSDI**, USENIX Symposium on Networked Systems Design and Implementation.
- 2025 **PPoPP**, ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming 2025.
- 2024 **SOCC**, ACM Symposium on Cloud Computing.
- 2024 **IEEE Cloud**, IEEE International Conference on Cloud Computing.
- 2024 **HPDC**, ACM International Symposium on High-Performance Parallel and Distributed Computing.
- 2024 **TCC**, Theory of Cryptography Conference.
- 2024 **IPDPS**, IEEE International Parallel & Distributed Processing Symposium.
- 2023 **SOCC**, ACM Symposium on Cloud Computing.
- 2023 **HotStorage**, ACM Workshop on Hot Topics in Storage and File Systems.
- 2023 **IEEE Cloud**, IEEE International Conference on Cloud Computing.
- 2023 **HPDC**, ACM International Symposium on High-Performance Parallel and Distributed Computing.
- 2023 **IPDPS**, IEEE International Parallel & Distributed Processing Symposium.
- 2022 **SEC**, ACM/IEEE Symposium on Edge Computing.
- 2022 **HPDC**, ACM International Symposium on High-Performance Parallel and Distributed Computing.
- 2021 **HPDC**, ACM International Symposium on High-Performance Parallel and Distributed Computing.

Relevant Academic Coursework

- Systems Operating Systems, Distributed Systems, Database Systems, Concurrent and Distributed Systems.
- AI/ML Large-Scale Optimization for Deep Learning, Deep Learning, Advanced Artificial Intelligence, Artificial Intelligence, Machine Learning, Data Mining.
- Core Data Structures, Computer Architecture, C Programming, Compilers, Object Oriented Software, Software Engineering, Algorithm Analysis, Graph Algorithms, Mobile Apps, Computer Networks.